

1 Solution — GIAM 3.1

Problem 6

1.1 Problem

Prove that for every real number x ,

$$\frac{2}{3} < x < \frac{3}{4} \implies \lfloor 12x \rfloor = 8.$$

1.2 Proof

Proof. Let x be a real number and suppose $\frac{2}{3} < x < \frac{3}{4}$. Multiplying through by 12 — which is positive, so the inequalities keep their direction — gives

$$12 \cdot \frac{2}{3} < 12x < 12 \cdot \frac{3}{4}, \text{ that is, } 8 < 12x < 9.$$

In particular $8 \leq 12x < 9$. By the definition of the floor function, $\lfloor t \rfloor = n$ exactly when $n \leq t < n + 1$. Taking $n = 8$ and $t = 12x$ gives $\lfloor 12x \rfloor = 8$. ■

1.3 The Picture

Scaling by 12 sends the interval $(2/3, 3/4)$ exactly onto $(8, 9)$, where the floor is pinned to 8.

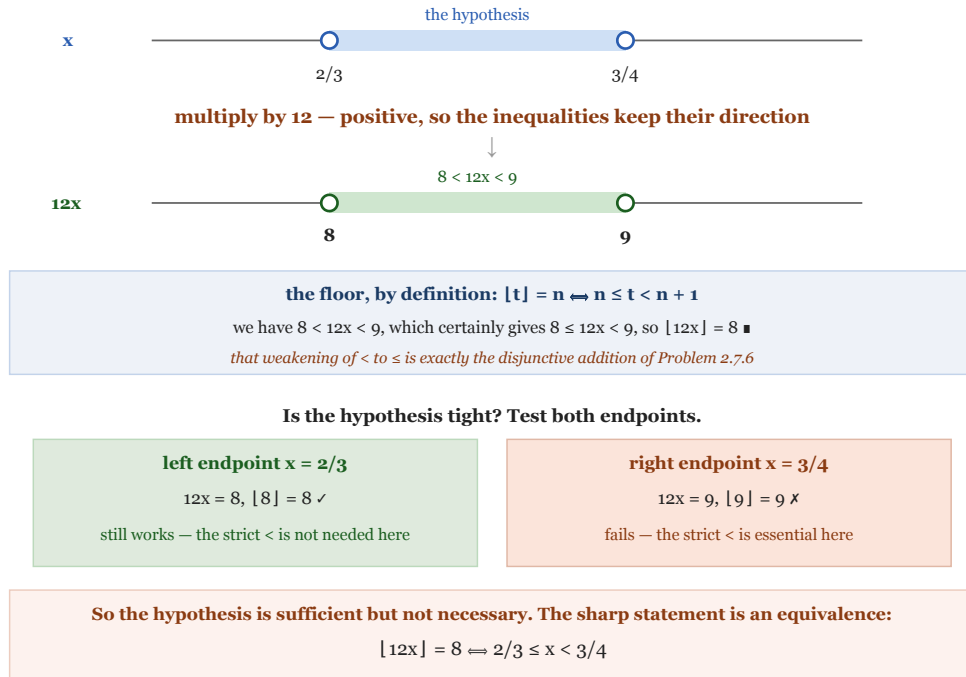


Figure 1.1: Top: a number line for x with the open interval from $2/3$ to $3/4$ shaded, both endpoints drawn as open circles, labelled ‘the hypothesis’. An arrow notes: multiply by 12 — positive, so the inequalities keep their direction. Below, a number line for $12x$ with the open interval from 8 to 9 shaded, labelled 8 less than $12x$ less than 9. A panel states the floor definition: the floor of t equals n exactly when n is at most t and t is less than n plus 1; we have 8 less than $12x$ less than 9, which certainly gives 8 at most $12x$ less than 9, so the floor of $12x$ is 8 — and notes that weakening the strict inequality to a non-strict one is exactly the disjunctive addition of Problem 2.7.6. Below, two endpoint tests. Left endpoint x equals $2/3$: $12x$ equals 8 and the floor of 8 is 8, so it still works and the strict inequality is not needed there. Right endpoint x equals $3/4$: $12x$ equals 9 and the floor of 9 is 9, so it fails and the strict inequality is essential there. A closing panel: the hypothesis is sufficient but not necessary, and the sharp statement is the equivalence — floor of $12x$ equals 8 if and only if $2/3$ is at most x and x is less than $3/4$.

The proof has exactly two moves: **scale the interval**, then **read off the floor**. Scaling by 12 is what makes the problem work, because $12 \cdot \frac{2}{3} = 8$ and $12 \cdot \frac{3}{4} = 9$ are *consecutive integers* — so the scaled interval sits inside one “floor step.”

Verified on 2000 random points of $(\frac{2}{3}, \frac{3}{4})$.

1.4 Remark — The Weakening Step Is Problem 2.7.6

There is a small logical move in the middle worth naming. We derive $8 < 12x$ but the floor characterisation requires $8 \leq 12x$, so we weaken the strict inequality to a non-strict one. That is precisely the **disjunctive addition** of Problem 2.7.6: since $a \leq b$ abbreviates $(a < b) \vee (a = b)$, inferring $8 \leq 12x$ from $8 < 12x$ is the inference $P \vdash P \vee Q$.

It costs nothing and is easy to perform silently, but it is exactly the situation that problem described — you weaken because the *next* tool demands a specific shape, not because the weaker fact is more informative.

1.5 Remark — Why the Multiplier Is 12

The number **12** is not arbitrary: it is $\text{lcm}(3, 4)$, the smallest positive integer that clears **both** denominators at once. Smaller multipliers fail to land both endpoints on integers:

multiplier m	$m \cdot \frac{2}{3}$	$m \cdot \frac{3}{4}$	both integers?
3	2	9/4	no
4	8/3	3	no
6	4	9/2	no
12	8	9	yes

Table 1.1

And landing on *consecutive* integers is what pins the floor to a single value. Had the endpoints scaled to, say, 8 and 11, the floor could be 8, 9 or 10 and no single answer would exist. So the problem is really engineered around the fact that $\frac{3}{4} - \frac{2}{3} = \frac{1}{12}$ — the interval is exactly one twelfth long, hence exactly one unit long after scaling.

1.6 Remark — The Hypothesis Is Not Tight on the Left

Testing the endpoints reveals an asymmetry:

- $x = \frac{2}{3}$: then $12x = 8$ and $\lfloor 8 \rfloor = 8$. The conclusion **still holds**, so the strict " $<$ " on the left was not needed.
- $x = \frac{3}{4}$: then $12x = 9$ and $\lfloor 9 \rfloor = 9 \neq 8$. The conclusion **fails**, so the strict " $<$ " on the right is essential.

The reason is that the floor function is **right-continuous** — it jumps *at* each integer and holds its new value on $[n, n+1)$ — so the left endpoint of each step belongs to it and the right endpoint does not. Consequently the sharp statement is a biconditional:

$$\lfloor 12x \rfloor = 8 \iff \frac{2}{3} \leq x < \frac{3}{4}.$$

(Verified on 20,000 random points, no mismatches.) The problem's hypothesis is therefore **sufficient but not necessary** — a perfectly legitimate way to state a theorem, but worth noticing.

1.7 Remark — Multiplying an Inequality Requires a Sign Check

The step “multiply through by 12” quietly uses that $12 > 0$. For a positive multiplier the order relation is preserved; for a negative one it **reverses**:

$$a < b \implies 12a < 12b, \quad \text{but} \quad a < b \implies -12a > -12b.$$

Here the multiplier is visibly positive so the check is instant, but the habit matters — forgetting to flip when multiplying by a negative is among the most common algebra errors, and in a proof it silently invalidates everything downstream. It is the same discipline as the closure checks in Problems 3.1.3 and 3.1.4: a step that “obviously works” still rests on a stated property.

1.8 Remark — The General Version

Nothing about this problem is special to 8, 12, or these fractions. The same two-step argument proves the general fact

$$\frac{n}{m} \leq x < \frac{n+1}{m} \iff \lfloor mx \rfloor = n \quad (m \text{ a positive integer}),$$

which says that multiplying by m and taking the floor is exactly the operation of asking “**which of the m equal subintervals of each unit does x fall into?**” That is why $\lfloor mx \rfloor$ appears throughout computer science — in hashing a real number into m

buckets, in converting a fraction to a fixed-point representation, and in the digit-extraction of Chapter 1's base conversions, where $\lfloor 10x \rfloor$ peels off one decimal digit at a time.